REFERENCES

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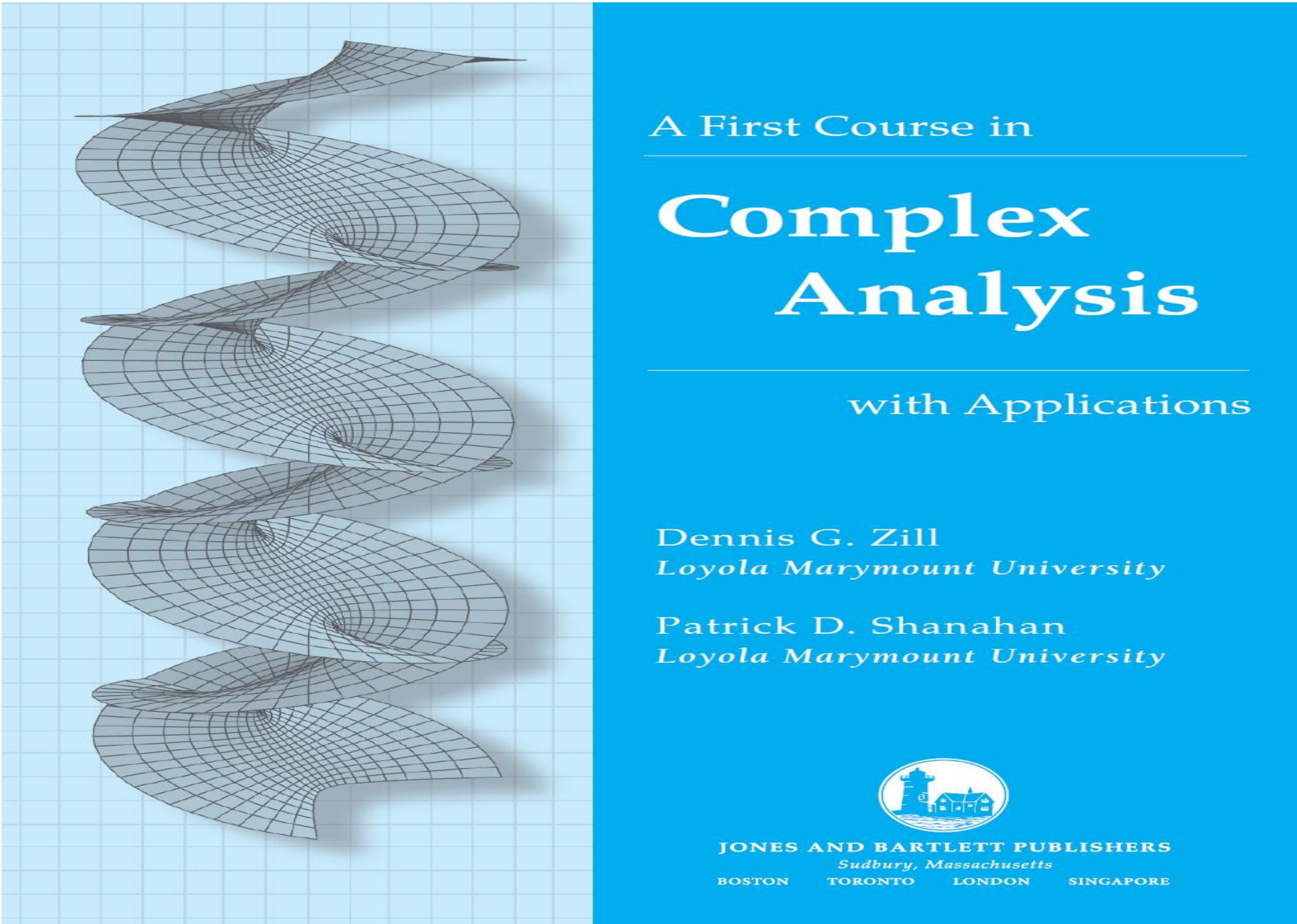

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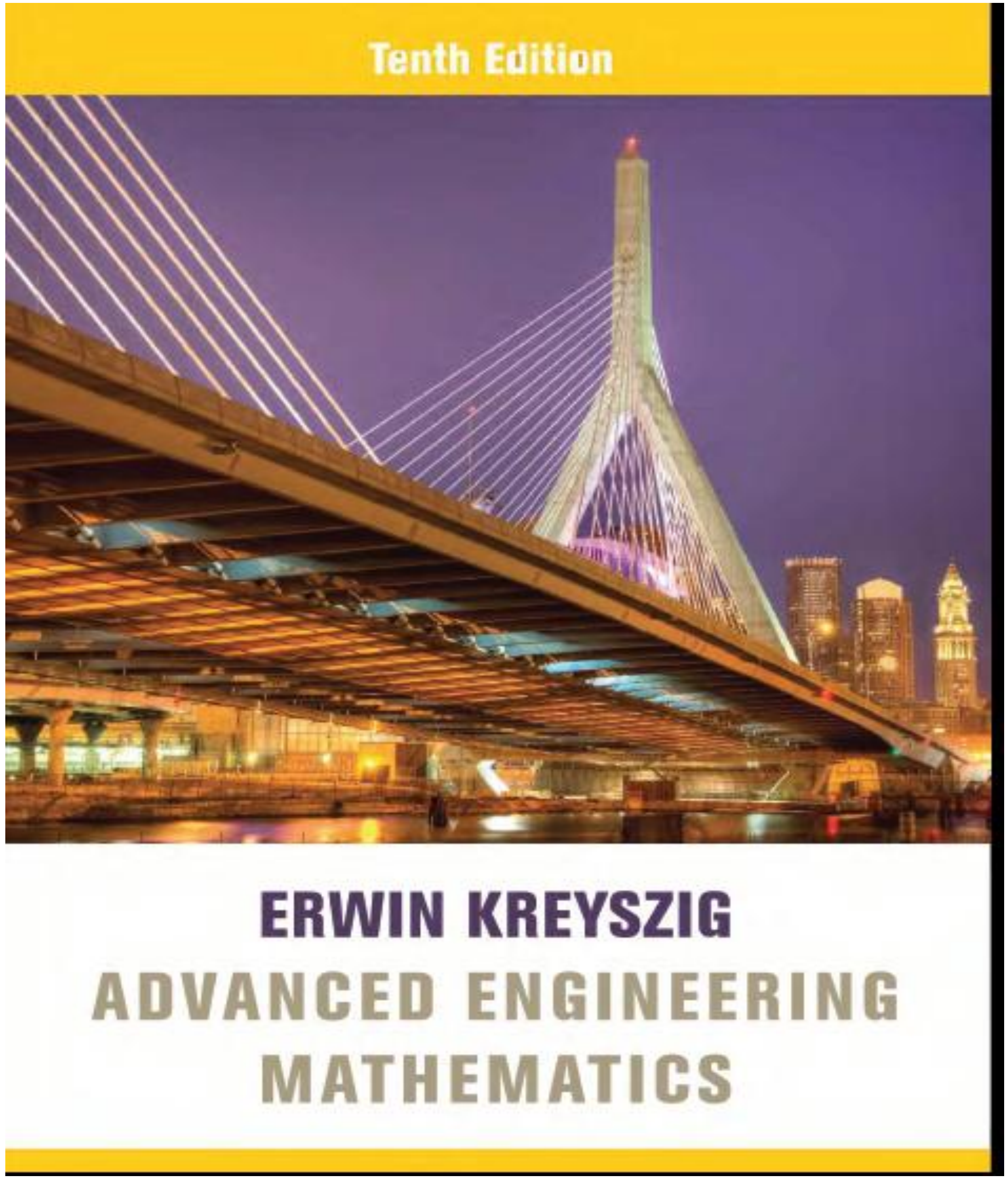
Tenth Edition
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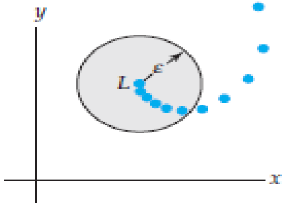
ADVANCED ENGINEERING
MATHEMATICS

Sequences A sequence $\{z_n\}$ is a function whose domain is the set of positive integers and whose range is a subset of the complex numbers C. In other words, to each integer $n = 1, 2, 3, \ldots$ we assign a single complex number z_n. For example, the sequence $\{1 + i^n\}$ is



If $\{z_n\}$ converges to L, all but a finite number of terms are in every ε-neighborhood of L.

$$\begin{array}{cccccc} 1+i, & 0, & 1-i, & 2, & 1+i, & \ldots . \\ \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \\ n=1, & n=2, & n=3, & n=4, & n=5, & \ldots \end{array} \tag{1}$$

If $\lim_{n\to\infty} z_n = L$, we say the sequence $\{z_n\}$ is **convergent**. In other words, $\{z_n\}$ converges to the number L if for each positive real number ε an N can be found such that $|z_n - L| < \varepsilon$ whenever $n > N$. Since $|z_n - L|$ is distance, the terms z_n of a sequence that converges to L can be made arbitrarily close to L. Put another way, when a sequence $\{z_n\}$ converges to L, then all but a finite number of the terms of the sequence are within every ε-neighborhood of L. A sequence that is not convergent is said to be **divergent**.

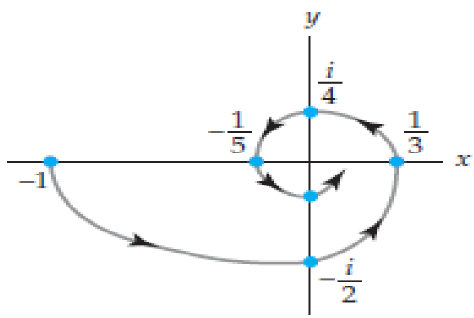
The sequence $\{1+i^n\}$ illustrated in (1) is divergent since the general term $z_n = 1 + i^n$ does not approach a fixed complex number as $n \to \infty$.

EXAMPLE 1 A Convergent Sequence

The sequence $\left\{\dfrac{i^{n+1}}{n}\right\}$ converges since $\lim\limits_{n\to\infty} \dfrac{i^{n+1}}{n} = 0$. As we see from

$$-1, -\frac{i}{2}, \frac{1}{3}, \frac{i}{4}, -\frac{1}{5}, \cdots,$$

and Figure , the terms of the sequence, marked by colored dots in the figure, spiral in toward the point $z = 0$ as n increases.



: The terms of the sequence $\{i^{n+1}/n\}$ spiral in toward 0.

Theorem Criterion for Convergence

A sequence $\{z_n\}$ converges to a complex number $L = a + ib$ if and only if $\mathrm{Re}(z_n)$ converges to $\mathrm{Re}(L) = a$ and $\mathrm{Im}(z_n)$ converges to $\mathrm{Im}(L) = b$.

EXAMPLE 2

Consider the sequence $\left\{ \dfrac{3+ni}{n+2ni} \right\}$. From

$$z_n = \frac{3+ni}{n+2ni} = \frac{(3+ni)(n-2ni)}{n^2+4n^2} = \frac{2n^2+3n}{5n^2} + i\frac{n^2-6n}{5n^2},$$

we see that

$$\mathrm{Re}(z_n) = \frac{2n^2+3n}{5n^2} = \frac{2}{5} + \frac{3}{5n} \to \frac{2}{5}$$

and $$\mathrm{Im}(z_n) = \frac{n^2-6n}{5n^2} = \frac{1}{5} - \frac{6}{5n} \to \frac{1}{5}$$

as $n \to \infty$. From Theorem , the last results are sufficient for us to conclude that the given sequence converges to $a + ib = \frac{2}{5} + \frac{1}{5}i$.

Series An infinite series or series of complex numbers

$$\sum_{k=1}^{\infty} z_k = z_1 + z_2 + z_3 + \cdots + z_n + \cdots$$

is **convergent** if the sequence of partial sums $\{S_n\}$, where

$$S_n = z_1 + z_2 + z_3 + \cdots + z_n$$

converges. If $S_n \to L$ as $n \to \infty$, we say that the series converges to L or that the sum of the series is L.

Geometric Series A **geometric series** is any series of the form

$$\sum_{k=1}^{\infty} az^{k-1} = a + az + az^2 + \cdots + az^{n-1} + \cdots. \tag{2}$$

For (2), the nth term of the sequence of partial sums is

$$S_n = a + az + az^2 + \cdots + az^{n-1}. \tag{3}$$

When an infinite series is a geometric series, it is always possible to find a formula for S_n. To demonstrate why this is so, we multiply S_n in (3) by z,

$$zS_n = az + az^2 + az^3 + \cdots + az^n,$$

and subtract this result from S_n. All terms cancel except the first term in S_n and the last term in zS_n:

$$\begin{aligned} S_n - zS_n &= \left(a + az + az^2 + \cdots + az^{n-1}\right) \\ &\quad - \left(az + az^2 + az^3 + \cdots + az^{n-1} + az^n\right) \\ &= a - az^n \end{aligned}$$

or $(1-z)S_n = a(1-z^n)$. Solving the last equation for S_n gives us

$$S_n = \frac{a(1-z^n)}{1-z}. \tag{4}$$

Now $z^n \to 0$ as $n \to \infty$ whenever $|z| < 1$, and so $S_n \to a/(1-z)$. In other words, for $|z| < 1$ the sum of a geometric series (2) is $a/(1-z)$:

$$\frac{a}{1-z} = a + az + az^2 + \cdots + az^{n-1} + \cdots. \tag{5}$$

A geometric series (2) diverges when $|z| \geq 1$.

If we set $a = 1$, the equality in (5) is

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots. \tag{6}$$

If we then replace the symbol z by $-z$ in (6), we get a similar result

$$\frac{1}{1+z} = 1 - z + z^2 - z^3 + \cdots. \tag{7}$$

Like (5), the equality in (7) is valid for $|z| < 1$ since $|-z| = |z|$. Now with $a = 1$, (4) gives us the sum of the first n terms of the series in (6):

$$\frac{1-z^n}{1-z} = 1 + z + z^2 + z^3 + \cdots + z^{n-1}.$$

If we rewrite the left side of the above equation as $\dfrac{1-z^n}{1-z} = \dfrac{1}{1-z} + \dfrac{-z^n}{1-z}$, we obtain an alternative form

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots + z^{n-1} + \frac{z^n}{1-z} \tag{8}$$

EXAMPLE 3 Convergent Geometric Series

The infinite series

$$\sum_{k=1}^{\infty} \frac{(1+2i)^k}{5^k} = \frac{1+2i}{5} + \frac{(1+2i)^2}{5^2} + \frac{(1+2i)^3}{5^3} + \cdots$$

is a geometric series. It has the form given in (2) with $a = \frac{1}{5}(1+2i)$ and $z = \frac{1}{5}(1+2i)$. Since $|z| = \sqrt{5}/5 < 1$, the series is convergent and its sum is given by (5):

$$\sum_{k=1}^{\infty} \frac{(1+2i)^k}{5^k} = \frac{\dfrac{1+2i}{5}}{1 - \dfrac{1+2i}{5}} = \frac{1+2i}{4-2i} = \frac{1}{2}i.$$

Theorem **A Necessary Condition for Convergence**

If $\sum_{k=1}^{\infty} z_k$ converges, then $\lim_{n\to\infty} z_n = 0$.

Theorem **The nth Term Test for Divergence**

If $\lim_{n\to\infty} z_n \neq 0$, then $\sum_{k=1}^{\infty} z_k$ diverges.

Definition [redacted] **Absolute and Conditional Convergence**

An infinite series $\sum_{k=1}^{\infty} z_k$ is said to be **absolutely convergent** if $\sum_{k=1}^{\infty} |z_k|$ converges. An infinite series $\sum_{k=1}^{\infty} z_k$ is said to be **conditionally convergent** if it converges but $\sum_{k=1}^{\infty} |z_k|$ diverges.

EXAMPLE 4 Absolute Convergence

The series $\sum_{k=1}^{\infty} \frac{i^k}{k^2}$ is absolutely convergent since the series $\sum_{k=1}^{\infty} \left|\frac{i^k}{k^2}\right|$ is the same as the real convergent p-series $\sum_{k=1}^{\infty} \frac{1}{k^2}$. Here we identify $p = 2 > 1$.

As in real calculus:

$$\sum_{k=1}^{\infty} \frac{i^k}{k^2} = i - \frac{1}{2^2} - \frac{i}{3^2} + \cdots$$

converges because it is shown to be absolutely convergent.

As in real calculus:

Absolute convergence implies convergence.

Theorem **Ratio Test**

Suppose $\sum_{k=1}^{\infty} z_k$ is a series of nonzero complex terms such that

$$\lim_{n\to\infty} \left| \frac{z_{n+1}}{z_n} \right| = L. \tag{9}$$

(*i*) If $L < 1$, then the series converges absolutely.

(*ii*) If $L > 1$ or $L = \infty$, then the series diverges.

(*iii*) If $L = 1$, the test is inconclusive.

Theorem Root Test

Suppose $\sum_{k=1}^{\infty} z_k$ is a series of complex terms such that

$$\lim_{n\to\infty} \sqrt[n]{|z_n|} = L. \tag{10}$$

(*i*) If $L < 1$, then the series converges absolutely.

(*ii*) If $L > 1$ or $L = \infty$, then the series diverges.

(*iii*) If $L = 1$, the test is inconclusive.

Power Series The notion of a power series is important in the study of analytic functions. An infinite series of the form

$$\sum_{k=0}^{\infty} a_k(z-z_0)^k = a_0 + a_1(z-z_0) + a_2(z-z_0)^2 + \cdots, \tag{11}$$

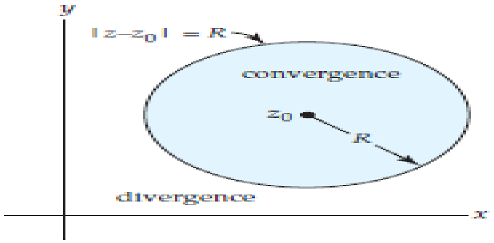
where the coefficients a_k are complex constants, is called a **power series in** $z - z_0$. The power series (11) is said to be **centered at** z_0; the complex point z_0 is referred to as the **center** of the series. In (11) it is also convenient to define $(z-z_0)^0 = 1$ even when $z = z_0$.

Circle of Convergence Every complex power series (11) has a **radius of convergence.** Analogous to the concept of an interval of convergence for real power series, a complex power series (11) has a **circle of convergence**, which is the circle centered at z_0 of largest radius $R > 0$ for which (11) converges at every point *within* the circle $|z - z_0| = R$. A power series converges absolutely at all points z within its circle of convergence, that is, for all z satisfying $|z - z_0| < R$, and diverges at all points z exterior to the circle, that is, for all z satisfying $|z - z_0| > R$. The radius of convergence can be:

(*i*) $R = 0$ (in which case (11) converges only at its center $z = z_0$),

(*ii*) R a finite positive number (in which case (11) converges at all interior points of the circle $|z - z_0| = R$), or

(*iii*) $R = \infty$ (in which case (11) converges for all z).



No general statement concerning convergence at points on the circle $|z - z_0| = R$ can be made.

EXAMPLE 5 Circle of Convergence

Consider the power series $\sum_{k=1}^{\infty} \frac{z^{k+1}}{k}$. By the ratio test (9),

$$\lim_{n\to\infty}\left|\frac{\frac{z^{n+2}}{n+1}}{\frac{z^{n+1}}{n}}\right| = \lim_{n\to\infty}\frac{n}{n+1}|z| = |z|.$$

Thus the series converges absolutely for $|z| < 1$. The circle of convergence is $|z| = 1$ and the radius of convergence is $R = 1$. Note that on the circle of convergence $|z| = 1$, the series does not converge absolutely since $\sum_{k=1}^{\infty} \frac{1}{k}$ is the well-known divergent harmonic series. Bear in mind *this does not say* that the series diverges on the circle of convergence. In fact, at $z = -1$, $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$ is the convergent alternating harmonic series. Indeed, it can be shown that the series converges at all points on the circle $|z| = 1$ *except* at $z = 1$.

IF

$$(i)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L \neq 0, \text{ the radius of convergence is } R = \frac{1}{L}; \tag{12}$$

$$(ii)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = 0, \text{ the radius of convergence is } R = \infty; \tag{13}$$

$$(iii)\ \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \infty, \text{ the radius of convergence is } R = 0. \tag{14}$$

Similar conclusions can be made for the root test (10) by utilizing

$$\lim_{n\to\infty} \sqrt[n]{|a_n|}. \tag{15}$$

For example, if $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L \neq 0$, then $R = 1/L$.

EXAMPLE 6 Radius of Convergence

Consider the power series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k!}(z-1-i)^k$. With the identification $a_n = (-1)^{n+1}/n!$ we have

$$\lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+2}}{(n+1)!}}{\frac{(-1)^{n+1}}{n!}} \right| = \lim_{n\to\infty} \frac{1}{n+1} = 0.$$

Hence by (13) the radius of convergence is ∞; the power series with center $z_0 = 1+i$ converges absolutely for all z, that is, for $|z-1-i| < \infty$.

EXAMPLE 7 Radius of Convergence

Consider the power series $\sum_{k=1}^{\infty} \left(\frac{6k+1}{2k+5}\right)^k (z-2i)^k$. With $a_n = \left(\frac{6n+1}{2n+5}\right)^n$, the root test in the form (15) gives

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{6n+1}{2n+5} = 3.$$

By reasoning similar to that leading to (12), we conclude that the radius of convergence of the series is $R = \frac{1}{3}$. The circle of convergence is $|z-2i| = \frac{1}{3}$; the power series converges absolutely for $|z-2i| < \frac{1}{3}$.

The Arithmetic of Power Series On occasion it may be to our advantage to perform certain *arithmetic operations* on one or more power series. Although it would take us too far afield to delve into properties of power series in a formal manner (stating and proving theorems), it will be helpful at points in this chapter to know what we can (or cannot) do to power series. So here are some facts.

- A power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ can be multiplied by a nonzero complex constant c without affecting its convergence or divergence.

- A power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ converges absolutely within its circle of convergence. As a consequence, within the circle of convergence the terms of the series can be rearranged and the rearranged series has the same sum L as the original series.

- Two power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ and $\sum_{k=0}^{\infty} b_k(z-z_0)^k$ can be added and subtracted by adding or subtracting like terms. In symbols:

$$\sum_{k=0}^{\infty} a_k(z-z_0)^k \pm \sum_{k=0}^{\infty} b_k(z-z_0)^k = \sum_{k=0}^{\infty} (a_k \pm b_k)(z-z_0)^k.$$

  If both series have the same nonzero radius R of convergence, the radius of convergence of $\sum_{k=0}^{\infty} (a_k \pm b_k)(z-z_0)^k$ is R. It should make intuitive sense that if one series has radius of convergence $r > 0$ and the other has radius of convergence $R > 0$, where $r \neq R$, then the radius of convergence of $\sum_{k=0}^{\infty} (a_k \pm b_k)(z-z_0)^k$ is the smaller of the two numbers r and R.

- Two power series can (with care) be multiplied and divided.

the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(2n)}{(n)^2}(z-3i)^n$ is

$$R = \lim_{n\to\infty}\left[\frac{(2n)}{(n)^2}\Big/\frac{(2n+2)}{((n+1))^2}\right] = \lim_{n\to\infty}\left[\frac{(2n)}{(2n+2)}\cdot\frac{((n+1))^2}{(n)^2}\right] = \lim_{n\to\infty}\frac{(n+1)^2}{(2n+2)(2n+1)} = \frac{1}{4}.$$

The series converges in the open disk $|z-3i| < \frac{1}{4}$ of radius $\frac{1}{4}$ and center $3i$.

Find the radius of convergence R of the power series

$$\sum_{n=0}^{\infty}\left[1+(-1)^n+\frac{1}{2^n}\right]z^n = 3+\frac{1}{2}z+\left(2+\frac{1}{4}\right)z^2+\frac{1}{8}z^3+\left(2+\frac{1}{16}\right)z^4+\cdots.$$

Solution.

It can be shown that

$$R = 1/\tilde{L}, \qquad \tilde{L} = \lim_{n\to\infty}\sqrt[n]{|a_n|}.$$

This still does not help here, since $(\sqrt[n]{|a_n|})$ does not converge because $\sqrt[n]{|a_n|} = \sqrt[n]{1/2^n} = \frac{1}{2}$ for odd n, whereas for even n we have

$$\sqrt[n]{|a_n|} = \sqrt[n]{2+1/2^n} \to 1 \qquad \text{as} \qquad n\to\infty,$$

so that $\sqrt[n]{|a_n|}$ has the two limit points $\frac{1}{2}$ and 1. It can further be shown that

$$R = 1/\tilde{l}, \qquad \tilde{l} \text{ the greatest limit point of the sequence } \left\{\sqrt[n]{|a_n|}\right\}.$$

Here $\tilde{l} = 1$, so that $R = 1$. *Answer.* The series converges for $|z| < 1$.

THEOREM

Termwise Differentiation of a Power Series

The derived series of a power series has the same radius of convergence as the original series.

The *geometric series*

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots$$

converges absolutely if $|z| < 1$ and diverges if $|z| \geqq 1$

Find the radius of convergence R of the following series by applying Theorem

$$\sum_{n=2}^{\infty} \binom{n}{2} z^n = z^2 + 3z^3 + 6z^4 + 10z^5 + \cdots .$$

Solution. Differentiate the geometric series twice term by term and multiply the result by $z^2/2$. This yields the given series. Hence $R = 1$ by Theorem

Termwise Integration of Power Series

The power series

$$\sum_{n=0}^{\infty} \frac{a_n}{n+1} z^{n+1} = a_0 z + \frac{a_1}{2} z^2 + \frac{a_2}{3} z^3 + \cdots$$

obtained by integrating the series $a_0 + a_1 z + a_2 z^2 + \cdots$ *term by term has the same radius of convergence as the original series.*

Taylor Series

The correspondence between a complex number z within the circle of convergence and the number to which the series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ converges is single-valued. In this sense, a power series *defines* or *represents* a function f; for a specified z within the circle of convergence, the number L to which the power series converges is defined to be the value of f at z, that is, $f(z) = L$. In this section we present some important facts about the nature of this function f.

In the preceding section we saw that every power series has a radius of convergence R. Throughout the discussion in this section we will assume that a power series $\sum_{k=0}^{\infty} a_k(z - z_0)^k$ has either a *positive* or an *infinite* radius R of convergence.

Theorem [redacted] Continuity

A power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ represents a continuous function f within its circle of convergence $|z-z_0| = R$.

Theorem [redacted] Term-by-Term Differentiation

A power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ can be differentiated term by term within its circle of convergence $|z-z_0| = R$.

Theorem [redacted] Term-by-Term Integration

A power series $\sum_{k=0}^{\infty} a_k(z-z_0)^k$ can be integrated term-by-term within its circle of convergence $|z-z_0| = R$, for every contour C lying entirely within the circle of convergence.

Taylor Series Suppose a power series represents a function f within $|z - z_0| = R$, that is,

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + a_3(z - z_0)^3 + \cdots . \quad (1)$$

It follows from Theorem that the derivatives of f are the series

$$f'(z) = \sum_{k=1}^{\infty} a_k k(z - z_0)^{k-1} = a_1 + 2a_2(z - z_0) + 3a_3(z - z_0)^2 + \cdots , \quad (2)$$

$$f''(z) = \sum_{k=2}^{\infty} a_k k(k-1)(z - z_0)^{k-2} = 2 \cdot 1a_2 + 3 \cdot 2a_3(z - z_0) + \cdots , \quad (3)$$

$$f'''(z) = \sum_{k=3}^{\infty} a_k k(k-1)(k-2)(z - z_0)^{k-3} = 3 \cdot 2 \cdot 1a_3 + \cdots , \quad (4)$$

and so on. Since the power series (1) represents a differentiable function f within its circle of convergence $|z - z_0| = R$, where R is either a positive number or infinity, we conclude that a *power series represents an analytic function* within its circle of convergence.

There is a relationship between the coefficients a_k in (1) and the derivatives of f. Evaluating (1), (2), (3), and (4) at $z = z_0$ gives

$$f(z_0) = a_0, \quad f'(z_0) = 1!a_1, \quad f''(z_0) = 2!a_2, \quad \text{and} \quad f'''(z_0) = 3!a_3,$$

respectively. In general, $f^{(n)}(z_0) = n!\,a_n$, or

$$a_n = \frac{f^{(n)}(z_0)}{n!}, \quad n \geq 0. \quad (5)$$

When $n = 0$ in (5), we interpret the zero-order derivative as $f(z_0)$ and $0! = 1$ so that the formula gives $a_0 = f(z_0)$. Substituting (5) into (1) yields

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k. \quad (6)$$

This series is called the **Taylor series** for f centered at z_0. A Taylor series with center $z_0 = 0$,

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k, \tag{7}$$

is referred to as a **Maclaurin series.**

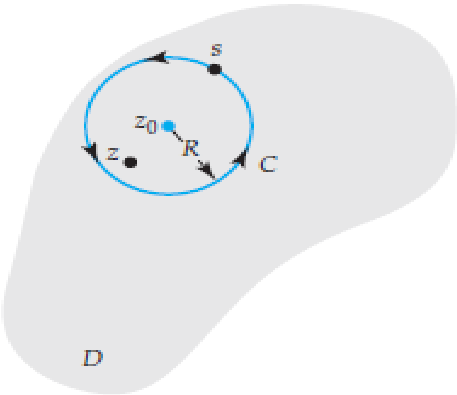
We have just seen that a power series with a nonzero radius R of convergence represents an analytic function. On the other hand we ask:

Question

If we are given a function f that is analytic in some domain D, can we represent it by a power series of the form (6) or (7)?

Theorem Taylor's Theorem

Let f be analytic within a domain D and let z_0 be a point in D. Then f has the series representation



$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!}(z - z_0)^k \tag{8}$$

valid for the largest circle C with center at z_0 and radius R that lies entirely within D.

Proof Let z be a fixed point within the circle C and let s denote the variable of integration. The circle C is then described by $|s - z_0| = R$. See Figure .
To begin, we use the Cauchy integral formula to obtain the value of f at z:

$$f(z) = \frac{1}{2\pi i}\oint_C \frac{f(s)}{s-z}ds = \frac{1}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)-(z-z_0)}ds$$

$$= \frac{1}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)}\left\{\frac{1}{1-\dfrac{z-z_0}{s-z_0}}\right\}ds. \tag{9}$$

By replacing z by $(z - z_0)/(s - z_0)$ in (8) , we have

$$\frac{1}{1-\dfrac{z-z_0}{s-z_0}} = 1 + \frac{z-z_0}{s-z_0} + \left(\frac{z-z_0}{s-z_0}\right)^2 + \cdots + \left(\frac{z-z_0}{s-z_0}\right)^{n-1} + \frac{(z-z_0)^n}{(s-z)(s-z_0)^{n-1}},$$

and so (9) becomes

$$\begin{aligned} f(z) = {} & \frac{1}{2\pi i}\oint_C \frac{f(s)}{s-z_0}ds + \frac{z-z_0}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^2}ds \\ & + \frac{(z-z_0)^2}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^3}ds + \cdots + \frac{(z-z_0)^{n-1}}{2\pi i}\oint_C \frac{f(s)}{(s-z_0)^n}ds \qquad (10) \\ & + \frac{(z-z_0)^n}{2\pi i}\oint_C \frac{f(s)}{(s-z)(s-z_0)^n}ds. \end{aligned}$$

Utilizing Cauchy's integral formula for derivatives, we can rewrite (10) as

$$f(z) = f(z_0) + \frac{f'(z_0)}{1!}(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots + \frac{f^{(n-1)}(z_0)}{(n-1)!}(z - z_0)^{n-1} + R_n(z), \quad (11)$$

where
$$R_n(z) = \frac{(z - z_0)^n}{2\pi i} \oint_C \frac{f(s)}{(s - z)(s - z_0)^n}\, ds.$$

Equation (11) is called Taylor's formula with remainder R_n. We now wish to show that the $R_n(z) \to 0$ as $n \to \infty$. This can be accomplished by showing that $|R_n(z)| \to 0$ as $n \to \infty$. Since f is analytic in D, we know that $|f(z)|$ has a maximum value M on the contour C. In addition, since z is inside C, $|z - z_0| < R$ and, consequently,

$$|s - z| = |s - z_0 - (z - z_0)| \geq |s - z_0| - |z - z_0| = R - d,$$

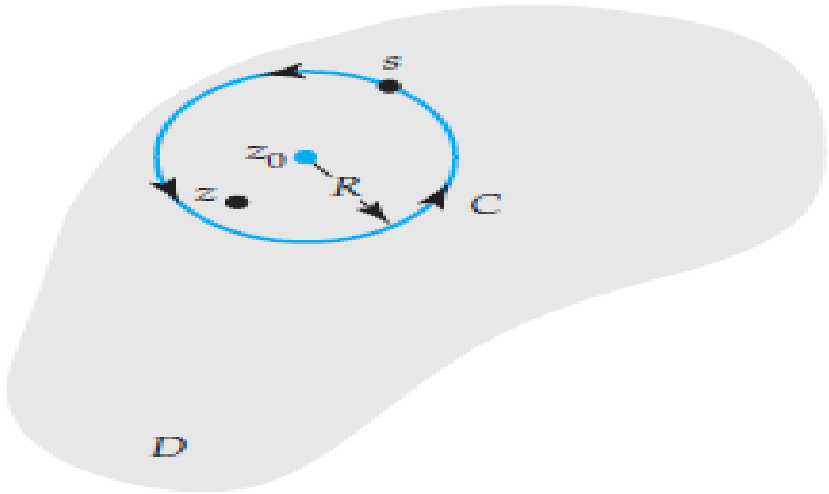
where $d = |z - z_0|$ is the distance from z to z_0. The *ML*-inequality then gives

$$|R_n(z)| = \left| \frac{(z - z_0)^n}{2\pi i} \oint_C \frac{f(s)}{(s - z)(s - z_0)^n}\, ds \right| \leq \frac{d^n}{2\pi} \cdot \frac{M}{(R - d)R^n} \cdot 2\pi R = \frac{MR}{R - d}\left(\frac{d}{R}\right)^n.$$

Because $d < R$, $(d/R)^n \to 0$ as $n \to \infty$ we conclude that $|R_n(z)| \to 0$ as $n \to \infty$. It follows that the infinite series

$$f(z_0) + \frac{f'(z_0)}{1!}(z - z_0) + \frac{f''(z_0)}{2!}(z - z_0)^2 + \cdots$$

converges to $f(z)$. In other words, the result in (8) is valid for any point z interior to C.



Some Important Maclaurin Series

$$e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \cdots = \sum_{k=0}^{\infty} \frac{z^k}{k!} \tag{12}$$

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{(2k+1)!} \tag{13}$$

$$\cos z = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k}}{(2k)!} \tag{14}$$

$$\mathrm{Ln}\,(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - + \cdots \qquad (|z| < 1).$$

Replacing z by $-z$ and multiplying both sides by -1, we get

$$-\mathrm{Ln}\,(1-z) = \mathrm{Ln}\frac{1}{1-z} = z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots \qquad (|z| < 1).$$

By adding both series we obtain

$$\mathrm{Ln}\frac{1+z}{1-z} = 2\left(z + \frac{z^3}{3} + \frac{z^5}{5} + \cdots\right) \qquad (|z| < 1).$$

Let $f(z) = 1/(1 - z)$. Then we have $f^{(n)}(z) = n/(1 - z)^{n+1}$, $f^{(n)}(0) = n$. Hence the Maclaurin expansion of $1/(1 - z)$ is the geometric series

* $$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots \qquad (|z| < 1).$$

$f(z)$ is singular at $z = 1$; this point lies on the circle of convergence.

Find the Maclaurin series of $f(z) = 1/(1 + z^2)$.

Solution. By substituting $-z^2$ for z in * we obtain

** $$\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = \sum_{n=0}^{\infty} (-z^2)^n = \sum_{n=0}^{\infty} (-1)^n z^{2n} = 1 - z^2 + z^4 - z^6 + \cdots \qquad (|z| < 1).$$

Find the Maclaurin series of $f(z) = \arctan z$.

Solution. We have $f'(z) = 1/(1 + z^2)$. Integrating ** term by term and using $f(0) = 0$ we get

$$\arctan z = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} z^{2n+1} = z - \frac{z^3}{3} + \frac{z^5}{5} - + \cdots \qquad (|z| < 1);$$

this series represents the principal value of $w = u + iv = \arctan z$ defined as that value for which $|u| < \pi/2$.

Develop $1/(c - z)$ in powers of $z - z_0$, where $c - z_0 \neq 0$.

Solution. This was done in the proof of Theorem 1, where $c = z$. The beginning was simple algebra and then the use of) with z replaced by $(z - z_0)/(c - z_0)$:

$$\frac{1}{c - z} = \frac{1}{c - z_0 - (z - z_0)} = \frac{1}{(c - z_0)\left(1 - \dfrac{z - z_0}{c - z_0}\right)} = \frac{1}{c - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{c - z_0}\right)^n$$

$$= \frac{1}{c - z_0}\left(1 + \frac{z - z_0}{c - z_0} + \left(\frac{z - z_0}{c - z_0}\right)^2 + \cdots\right).$$

This series converges for

$$\left|\frac{z - z_0}{c - z_0}\right| < 1, \qquad \text{that is,} \qquad |z - z_0| < |c - z_0|.$$

EXAMPLE 1 Radius of Convergence

Suppose the function $f(z) = \dfrac{3-i}{1-i+z}$ is expanded in a Taylor series with center $z_0 = 4-2i$. What is its radius of convergence R?

Solution Observe that the function is analytic at every point except at $z = -1+i$, which is an isolated singularity of f. The distance from $z = -1+i$ to $z_0 = 4-2i$ is

$$|z - z_0| = \sqrt{(-1-4)^2 + (1-(-2))^2} = \sqrt{34}.$$

This last number is the radius of convergence R for the Taylor series centered at $4-2i$.

EXAMPLE 2 **Maclaurin Series**

Find the Maclaurin expansion of $f(z) = \dfrac{1}{(1-z)^2}$.

Solution We could, of course, begin by computing the coefficients using (8). However, recall that for $|z| < 1$,

$$\frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots . \tag{15}$$

If we differentiate both sides of the last result with respect to z, then

$$\frac{d}{dz}\frac{1}{1-z} = \frac{d}{dz}1 + \frac{d}{dz}z + \frac{d}{dz}z^2 + \frac{d}{dz}z^3 + \cdots$$

or

$$\frac{1}{(1-z)^2} = 0 + 1 + 2z + 3z^2 + \cdots = \sum_{k=1}^{\infty} k\, z^{k-1}. \tag{16}$$

the radius of convergence of the last power series is the same as the original series, $R = 1$.

EXAMPLE 3 Taylor Series

Expand $f(z) = \dfrac{1}{1-z}$ in a Taylor series with center $z_0 = 2i$.

Solution In this solution we again use the geometric series (15). By adding and subtracting $2i$ in the denominator of $1/(1-z)$, we can write

$$\frac{1}{1-z} = \frac{1}{1-z+2i-2i} = \frac{1}{1-2i-(z-2i)} = \frac{1}{1-2i}\frac{1}{1-\dfrac{z-2i}{1-2i}}.$$

We now write $\dfrac{1}{1-\dfrac{z-2i}{1-2i}}$ as a power series by using (15) with the symbol z replaced by the expression $\dfrac{z-2i}{1-2i}$:

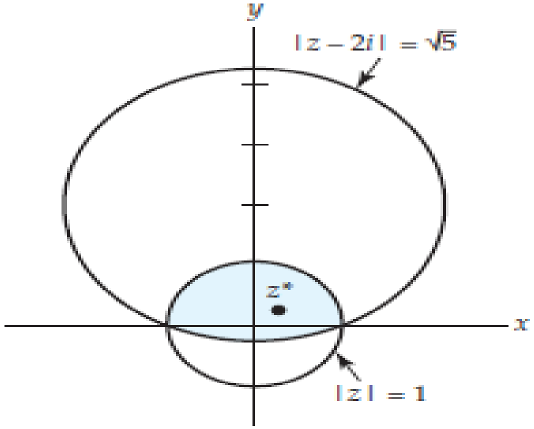
$$\frac{1}{1-z} = \frac{1}{1-2i}\left[1 + \frac{z-2i}{1-2i} + \left(\frac{z-2i}{1-2i}\right)^2 + \left(\frac{z-2i}{1-2i}\right)^3 + \cdots\right]$$

or

$$\frac{1}{1-z} = \frac{1}{1-2i} + \frac{1}{(1-2i)^2}(z-2i) + \frac{1}{(1-2i)^3}(z-2i)^2 + \frac{1}{(1-2i)^4}(z-2i)^3 + \cdots. \quad (17)$$

Because the distance from the center $z_0 = 2i$ to the nearest singularity $z = 1$ is $\sqrt{5}$, we conclude that the circle of convergence for (17) is $|z-2i| = \sqrt{5}$. This can be verified by the ratio test of the preceding section.

In (15) and (17) we represented the same function $f(z) = 1/(1-z)$ by two different power series. The first series (15) has center $z_0 = 0$ and radius of convergence $R = 1$. The second series (17) has center $z_0 = 2i$ and radius of convergence $R = \sqrt{5}$. The two different circles of convergence are illustrated in Figure   The interior of the intersection of the two circles, shown in color, is the region where *both* series converge; in other words, at a specified point z^* in this region, both series converge to same value $f(z^*) = 1/(1-z^*)$.



Series (15) and (17) both converge in the shaded region.